NASTARAN ESMAEIL ZADEH

Aspiring AI Engineer | Fresh Graduate | Python | Machine Learning | NLP | Generative AI

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Science fresh graduate and MSc Artificial Intelligence student with project experience in machine learning, NLP, and AI chatbot development. Skilled in Python, data preprocessing, feature engineering, and model evaluation, with experience building web applications using React, Supabase, and MySQL. Seeking an entry-level AI engineering role to apply practical programming and problem-solving skills.

PROFESSIONAL EXPERIENCE

Founder & President (Part-time)

Iranian Cultural Club (ICC), Asia Pacific University | Kuala Lumpur, Malaysia | On-site **August 2024 - August 2026**

- Founded the Iranian Cultural Club to promote cultural exchange and strengthen the Iranian student community at APU.
- Led the executive committee and coordinated club operations, strategic planning, and member engagement.
- Planned cultural, educational, and networking events with university departments and student organizations.
- Managed event logistics, budgeting, sponsorship opportunities, team coordination, and stakeholder relationships.

Web Development Intern

RA Marketing MY | Kuala Lumpur, Malaysia | On-site **January 2025 - May 2025**

- Developed and maintained responsive websites using WordPress, Drupal, HTML, and CSS.
- Designed and developed the SEO Consultants service page for Rose Attractions: roseattractions.com/our-services/seo-consultants.
- Contributed to the Rose Attractions homepage and built responsive landing pages focused on UI/UX, usability, and performance.
- Set up Google Analytics and performed on-page SEO improvements covering metadata, content structure, and usability.
- Conducted keyword research and competitor analysis while collaborating with the marketing team to strengthen digital presence.

EDUCATION

Master of Science in Artificial Intelligence

Asia Pacific University of Technology & Innovation (APU), Kuala Lumpur, Malaysia **May 2026 - Present**

Bachelor of Science (Hons) in Computer Science

Asia Pacific University of Technology & Innovation (APU), Kuala Lumpur, Malaysia **March 2023 - February 2026**

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB, SQL, JavaScript, TypeScript, HTML, CSS

Machine Learning & Data: Pandas, NumPy, Scikit-learn, data preprocessing, feature engineering, SMOTE, model training, model evaluation, deep learning, NLP, fuzzy logic, classical computer vision

AI & Content Creation: Agentic AI, generative AI, AI-assisted development, prompt engineering, AI chatbot development, AI-assisted video editing, Pictory AI, script-to-video production

Web, Database & SEO: Full-stack web development, React, Supabase, MySQL, Streamlit, WordPress, Drupal, responsive web design, UI/UX design, Google Analytics, on-page SEO, keyword research

Tools & Design: Git, GitHub, Jupyter Notebook, VS Code, Figma, Canva

Research & Professional: Research writing, analytical thinking, critical thinking, problem-solving, leadership, project management, team collaboration, communication

RESEARCH & PUBLICATIONS

Jokanian, S., & Esmaeil Zadeh, N. (2026). *Overcoming Bushfire Challenges via Machine Learning Techniques and Enhanced Training Data in Australia.* Manuscript under review at *Natural Hazards*. Manuscript No. NHAZ-D-25-04544.

- Co-authored research integrating satellite, drone, sensor, weather, and historical fire data with CNN and other machine learning methods for bushfire prediction and real-time monitoring.
- Evaluated a bushfire detection model using a 70% training, 20% validation, and 10% testing split, reporting 86% accuracy while examining data-transmission and environmental limitations.

PROJECTS

Hospital Readmission Prediction Using Machine Learning
Python | Pandas | Scikit-learn | SMOTE | Machine Learning
- Developed an end-to-end machine learning pipeline using the UCI Diabetes 130-US Hospitals dataset to predict patient readmission within 30 days.
- Addressed class imbalance using SMOTE and evaluated Logistic Regression, Decision Tree, Random Forest, Linear SVM, and MLP models using F1-score and ROC-AUC.

GlucoGuide - AI-Powered Diabetes Education Chatbot
React | TypeScript | TanStack | Supabase | Generative AI
- Developed a full-stack AI chatbot that provides structured diabetes education, safety guidance, learning takeaways, sources, and suggested follow-up questions.
- Implemented secure authentication, synchronized conversations, learning-progress tracking, user settings, and Row-Level Security using Supabase.

ICU Admission Decision Support Using Fuzzy Logic
Python | Scikit-Fuzzy | Fuzzy Inference System
- Developed a fuzzy decision-support system that evaluates SpO2, blood pressure, GCS, and heart rate to determine ICU admission priority.
- Designed 16 membership functions, clinical decision rules, fuzzification, inference, and defuzzification components with structured test cases.

University Professional Attire Identification
MATLAB | Image Processing | Computer Vision
- Designed a classical computer-vision pipeline to identify professional and non-professional university attire without deep learning.
- Applied image preprocessing, illumination correction, body-region analysis, and color, texture, and edge feature extraction.

InsuSmart - Web-Based Insulin Management System
Python | MySQL | HTML | CSS | JavaScript | UI/UX
- Developed key features for a web-based insulin inventory and management system using Python and MySQL.
- Designed database structures, dashboards, user workflows, and AI-assisted chatbot prompts using Figma and structured prompt engineering.

Airline Tweet Sentiment Classification Using Deep Learning
Python | Deep Learning | Natural Language Processing
- Developed a deep-learning sentiment-analysis workflow to classify airline tweets as positive, negative, or neutral.
- Preprocessed textual data and organized the training and evaluation workflow in a reproducible Jupyter Notebook project.

CERTIFICATIONS

AI Agents: From Zero to Production - Workshop 1
APU Hackthletes, Asia Pacific University **July 2026**
Certificate of Participation

Overreliance of AI in Education: Balancing Technology and Humanity for Quality Learning
Asia Pacific University **November 2025**
Certificate of Participation - Webinar

Web Design Pack Certificate **December 2022**
Tehran Institute of Technology | Credential ID: 1421266

VOLUNTEERING

Volunteer Food Sorter
The Lost Food Project | Kuala Lumpur, Malaysia **August 2024 - August 2025**
- Inspected and sorted donated produce based on freshness and suitability for safe distribution.
- Prepared edible food for partner charities, contributing to food-waste reduction and community support.

LANGUAGES

Persian: Native | **English:** Professional Working Proficiency | **Turkish:** Beginner

REFERENCES

Shahab Alizadeh
Senior Lecturer | shahab.alizadeh@apu.edu.my

Vijayaraj C. Vijayasingam
Senior Lecturer | vijayaraj@apu.edu.my